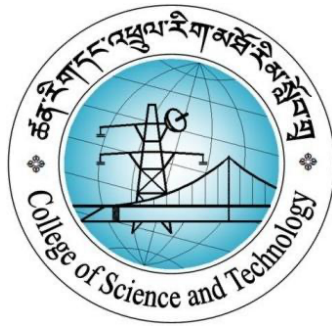


SWE 101 Report



Second Year Semester I

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3rd MAY 2026.

1. Introduction

This project involves developing a mobile application called Study Task Manager using React Native with Expo. The main goal of creating this app was to improve my understanding of mobile app development, including UI design, navigation, state management, and animations.

The app helps users manage subjects, track tasks, and monitor progress. It is designed to be simple, user-friendly, and responsive, with a clean structure that can be easily improved in future.

2. Objective of the Project

The objectives of this project are:

- To understand the basic structure of a React Native application
- To develop a mobile app using React Native with Expo
- To use core components such as View, Text, ScrollView, and Touchable elements
- To implement navigation between multiple screens
- To manage and display data such as subjects and tasks
- To design a clean and user-friendly interface using proper styling
- To add animations for better user experience
- To make the application responsive on different devices

3. App Idea

This project is a **mobile application called Study Task Manager** developed using React Native with Expo. The app is designed to help students organize their academic tasks, manage subjects, and track their progress in a simple and effective way. It also helped me understand key concepts of mobile app development such as UI design, navigation, and animations.

4. Features

- View and manage different subjects
- Add and track tasks with due dates and priority levels
- Mark tasks as completed
- Display progress using visual progress bars
- Show user statistics like streaks and completed tasks
- Light and Dark mode support
- Simple, clean, and user-friendly interface

5. Navigation Flow

- **Home Screen** → Displays overview of tasks and progress
- **Categories Screen** → Shows list of subjects
- **Detail Screen** → Displays subject tasks or task details
- **Profile Screen** → Shows user info, stats, and settings

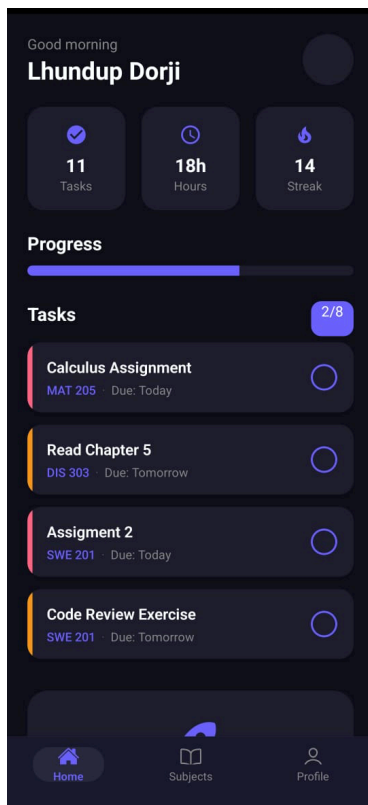
Navigation is implemented using multiple screens with smooth transitions between them.

6. Animations Used

- **Fade Animation** → Elements appear smoothly on screen
- **Slide Animation** → Components slide into view
- **Scale (Bounce) Animation** → Cards and avatars pop in with a bounce effect
- **Press Animation** → Buttons and rows slightly shrink when pressed
- **Animated Progress Bar** → Progress fills smoothly
- **Stagger Animation** → List items appear one after another

7. Screenshots

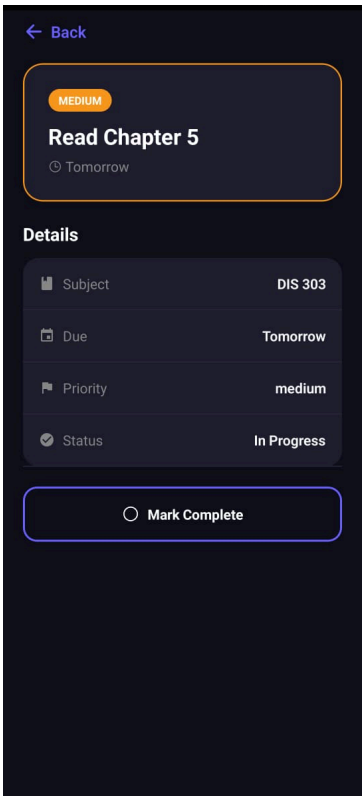
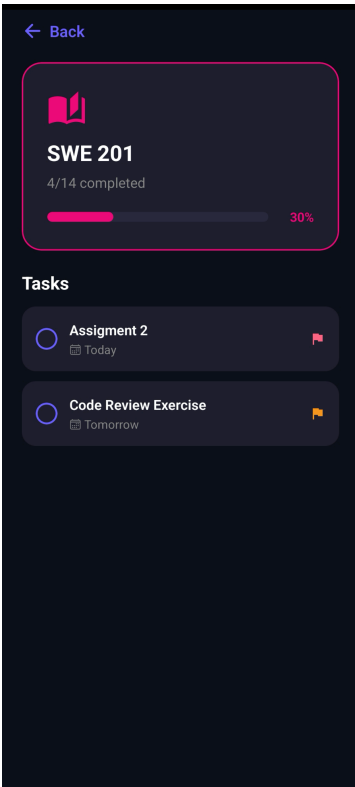
- **Home Screen**



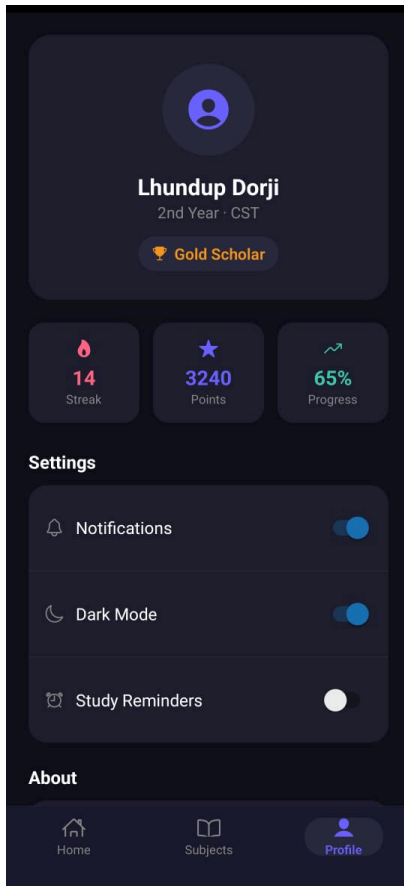
- Categories Screen



-
- Detail Screen



- **Profile Screen**



8. Challenges Faced

During the development of this project, a few challenges were encountered:

- Understanding how to implement navigation between multiple screens using React Navigation
- Managing component states and updating task data correctly
- Applying animations smoothly without affecting performance
- Organizing files and components into a clean and structured folder system

These challenges were overcome through practice, debugging, and testing different approaches, which helped improve overall understanding of React Native development.

9. Conclusion

This project demonstrates a simple yet effective student task management app with modern UI design, smooth navigation, and interactive animations, while helping build practical skills in React Native development.